

Algorithm Design of Bank Credit High Risk Customer Identification Based on Random Forest

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Abstract—The banking industry is a high-risk industry. With the continuous expansion of business areas and business scale, bank credit risk has become the most important risk. In this paper, a credit risk assessment model based on random forest (RF) algorithm is proposed, and the initial RF is optimized by parameter optimization, which further improves the recognition accuracy of the model. The experimental results show that, after many iterations, the accuracy of this method is obviously better than that of traditional support vector machine (SVM) algorithm, with an accuracy of 95.73%, 19.24% higher than that of SVM algorithm, and an error reduction of 30.26%. It can be seen that the credit risk assessment model based on RF algorithm is effective in identifying high-risk customers of bank credit, and it can provide some theoretical support for identifying high-risk customers of bank credit. Constructing the credit risk identification system of commercial banks with the help of modern intelligent algorithms can change the situation of superficial credit risk judgment and lagging risk response under the traditional management mode, improve the technical content and accuracy of risk analysis, and is of great significance for promoting commercial banks to improve their risk management level and realizing the stability and development of the financial industry.

Keywords—Commercial banks, High-risk customers, Credit risk; Random forest, Risk identification

I. INTRODUCTION

The banking industry is a high-risk industry. With the continuous expansion of business areas and business scale, bank credit risk has become the most important risk [1]. Therefore, the establishment of credit risk identification mechanism, effective control and reduction of credit risk have become one of the primary tasks of commercial banks. As an important part of the financial market, in the process of economic development, commercial banks not only undertake the basic duties of taking deposits and issuing loans, but also play a role in correctly guiding the flow of money, improving the efficiency of capital use and regulating the demand for funds in society, which has become a key factor in the development of national economy [2]. As the group customers of commercial banks, enterprises have the advantages of large-scale assets, strong capital strength, strong market competitiveness, wide financing channels, and low cost of mass marketing. They are favored by commercial banks and become the focus of many commercial banks [3]. However, through communication with bank practitioners and economic crime investigators, I found that many group customers have imperfect corporate governance and modern enterprise system, imperfect internal control measures, and failed investment decisions, resulting in the break of capital chain [4]. In recent years, the characteristics of group customers

have changed greatly, and the relationship between property rights within group customers is complex, multi-layered, natural person holding shares, proxy holding shares, etc., which makes it more difficult to identify the relationship[5]. As the main body and supplement of China's banking industry, commercial banks play an important role in promoting China's economic development.

Non-financial institutions are constantly deepening their financial innovation, and various financial derivatives are constantly introduced, which makes the living space of commercial banks in China suffer a very serious threat, and the traditional credit risk of commercial banks is further aggravated [6]. Individual group customers take advantage of the pressure of the bank's performance appraisal and competition in the same industry, and through multi-credit and over-credit, they force banks to compete to lower credit conditions, relax guarantee measures, and break through regulatory requirements, resulting in the inability to implement bank risk control measures [7]. There are many potential new risks in granting credit to group customers of commercial banks, which have the characteristics of strong concealment, great difficulty in bank supervision and inability to be touched by the existing risk control means [8]. It is of great practical significance to build a risk early warning system with China characteristics and international standards, and to establish a credit risk identification model and early warning mechanism that is in line with China's national conditions and has strong operability, which has become the main reason why this paper is going to carry out the research on credit risk identification of commercial banks [9]. In this paper, a credit risk assessment model based on RF algorithm is proposed, and the initial RF is optimized by parameter optimization, which further improves the recognition accuracy of the model.

II. METHODOLOGY

A. Analysis framework of credit risk data of commercial banks

Compared with individual customers, group customers bring greater economic benefits to banks, and at the same time, because the credit lines of group loans are relatively large, once risks occur, they also bring greater losses than individual customers. Group customers also have many drawbacks that can't be ignored in their management, which leads to a high concentration of risks in the concentrated capital investment of banks, and some of them have caused huge factual risks to commercial banks. Because of the complexity of related party transactions within the group customers, the credit rating of commercial banks may be distorted when investigating and reviewing the financial situation of one of the group customers. Some enterprises even deliberately fail to provide true and complete

information to commercial banks, which leads commercial banks to make wrong credit decisions based on these wrong and incomplete information [10]. According to complexity science, the complex system is in an unbalanced state, obeying the dynamic evolutionary theory, and under certain conditions, self-organizing phenomenon occurs by constantly exchanging energy, materials and information with the environment. However, according to the idea of

management entropy, any kind of management organization, system, policy, method, etc., always presents an irreversible process in which the effective energy gradually decreases and the ineffective energy continuously increases during the relatively closed organizational movement. The analysis framework of bank credit high-risk data based on RF algorithm is shown in Figure 1.

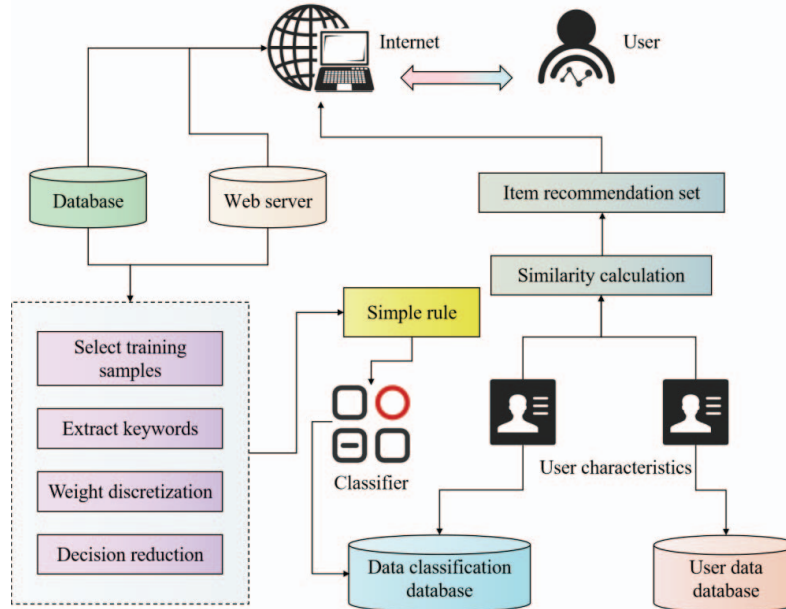


Figure 1. Analysis framework of bank credit high-risk data based on RF algorithm

For the information of cross-regional group customers, there is a lack of normal communication. Even if the information is received from other banks or the outside world, it is difficult to make an overall scientific assessment of group customers because of the lack of corresponding database to analyze, screen, classify and store the received information, and the overall coordination of information processing is poor. Group customers also take advantage of the defects of commercial banks and their branches in mastering customer information, and adopt multi-head financing and cross-loans in a number of commercial banks or branches of a commercial bank, which eventually leads to total over-financing. The corporate governance activities of group clients exceed the corporate boundary of enterprises. The corporate governance mechanism boundary of group clients is the boundary based on control rights, and the parent company influences and decides the decisions of subsidiaries by controlling their resources [11]. In real economic life, due to the complex property rights relationship of group customers, there is a complex double-layer and multi-layer principal-agent relationship. The more complex the agency chain, the lower the information transparency, and the behavior of subsidiaries may deviate from the will of the parent company. Because of the corporate governance boundary of group customers, the parent company is responsible for the behavior of subsidiaries. The mutual serial guarantee or repeated mortgage of assets between affiliated enterprises within the group customers leads to nominal secured loans becoming credit loans, and the guarantee is illusory. Many group enterprises have a low level of corporate governance and

internal control. In financial management, it is impossible to distinguish responsibilities and establish the necessary firewall. Once the borrower fails to repay the debt, it is easy to cause a chain reaction, which makes the whole group in trouble and makes it difficult to guarantee the bank's creditor's rights.

B. Algorithm for identifying high-risk customers in bank credit

Identifying and measuring risks exists in the whole business process, that is, first of all, we must have the idea and awareness of identifying risk points; secondly, how to quantify risks according to the discovered risk points; and finally, according to the measured risk degree, what kind of control measures should be taken. The credit risk identification model quantitatively describes the credit risks of different institutions, products, customers and customer groups, explores and excavates the factors and rules that affect the risks, and establishes a credit risk identification system of the whole bank, which can be effectively used for pre-lending, in-lending and post-lending management, so as to link the credit risk management before and after, and implement continuous tracking and monitoring for customers. Under the condition of sufficient market, systemic risk accounts for one of all risks faced by economic individuals. The systematic risks of customers mainly come from industry credit risks and regional credit risks, while the non-systematic risks mainly come from customers' business risks and financial risks. Therefore, only studying systematic risks or non-systematic risks of customers can't make a correct assessment of the risks faced

by customers, but should adopt a risk management model that combines systematic risks and non-systematic risks.

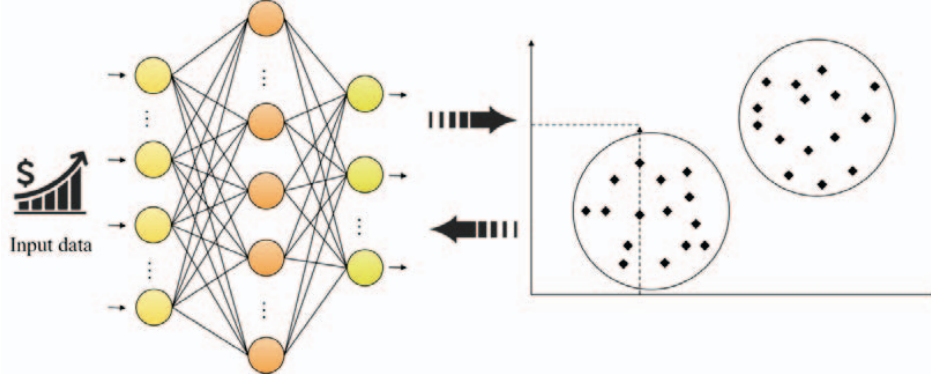


Figure 2. Cluster analysis of credit risk identification

To establish a risk early warning model, we can first study the relationship between several different types of variables and credit risks from several independent angles, and construct a relatively independent risk early warning subsystem. Then, considering the correlation between subsystems, the system is placed on the common platform of information collection, processing and application, and becomes a complete credit risk identification system, which carries out comprehensive analysis, processing and processing of various information to form the final risk warning [12]. In risk assessment, we should first determine the systemic risk of the customer, that is, study the credit risk of the industry and region to which it belongs, and preliminarily determine the systemic risk of the customer. Then, it analyzes the customer's management characteristics and financial risks, and determines the customer's non-systematic risks. Finally, the systematic risk and non-systematic risk of customers are comprehensively analyzed to determine the risk degree of customers.

The ability of a classifier to correctly classify data outside the training set is called generalization, which is the goal of each learning algorithm to optimize. Generalization error is the misclassification rate of the classifier to the data outside the training set. m assessment indexes make up the assessment values of n schemes. The assessment of n schemes by each assessment index can be expressed by index characteristic quantity, namely:

$$X = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \dots & \dots & \dots & \dots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{bmatrix} = (X_{ij})_{m \times n} \quad (1)$$

Where X_{ij} ($i = 1, 2, \dots, m; j = 1, 2, \dots, n$) is the index characteristic quantity of the i assessment factor of the j scheme. Calculate the set-valued statistical value x_{ij} of the i index of the j scheme with the following formula, namely:

The cluster analysis process of credit risk identification is shown in Figure 2.

$$X_{ij} = \frac{1}{2} \sum_{k=1}^n \left[\left(x_{2ij}^{(k)} \right)^2 - \left(x_{1ij}^{(k)} \right)^2 \right] / \sum_{k=1}^n \left[\left(x_{2ij}^{(k)} \right) - \left(x_{1ij}^{(k)} \right) \right] \quad (2)$$

Define the intensity of RF as the expectation of edge function:

$$s = E(mr(X, Y)) \quad (3)$$

The strength of the RF classifier is estimated as:

$$s = \frac{1}{n} \sum_{k=1}^n \left[Q(X_i, Y) - \max_{\substack{j \neq Y \\ i=1}} Q(X_i, j) \right] \quad (4)$$

Then the correlation between decision trees is:

$$\bar{\rho} = \frac{\text{var}(mr)}{sd(h(*)^2)} = \frac{\frac{1}{n} \sum_{k=1}^n \left[Q(X_i, Y) - \max_{\substack{j \neq Y \\ i=1}} Q(X_i, j) \right]^2 - s^2}{\left(\frac{1}{k} \sum_{u=1}^k \sqrt{p_u + \bar{p}_u} + (p_u - \bar{p}_u)^2 \right)^2} \quad (5)$$

$$p_u = \frac{\sum_{(x_i, y) \in O_i} I(h(x_i) = y)}{\sum_{(x_i, y) \in O_i} I(h(x_i))} \quad (6)$$

$$\bar{p}_u = \frac{\sum_{(x_i, y) \in O_i} I(h(x_i) = y_j)}{\sum_{(x_i, y) \in O_i} I(h(x_i))} \quad (7)$$

Where p_u represents the correct classification ratio of data samples outside the bag; $\bar{p}_u$ indicates the proportion

of y_j in the data samples out of the bag that are wrongly classified into multiple categories. $sd(h(*))^2$ is the square of the standard deviation of RF classifier, and $\text{var}(mr)$ is the variance of edge function.

Credit risk is influenced by national macro-economic policies, industry or regional economic development, micro-factors such as changes in customers' operating conditions, and meso-market factors. There are certain differences in the influence surface, degree and essence of various credit risk indicators on risk. Quantitatively describe the credit risks of the whole customers, industry customers and individual customers, explore and excavate the factors and rules that affect the risks, establish the credit risk identification system of the whole bank, make reasonable and timely decisions on the bank's credit business, strengthen the management of credit funds, prevent credit risks and ensure the safe and stable operation of the banking system.

III. RESULT ANALYSIS AND DISCUSSION

According to the existing industry experience, the whole sample data set is divided into training set and testing set, in which the training set is 60% of the original sample data set and the testing set is the remaining 40% of the original sample data set. The description and analysis of data characteristics gives the general distribution of bank customers, which belongs to qualitative analysis of attribute data. However, there are many attributes of bank data, and it is difficult to find important attributes only by qualitative analysis. Although the credit risk of commercial banks is caused by many uncertain factors, it doesn't mean that they can't cope with the credit risk. Commercial banks themselves and related personnel can analyze and control the credit risk, predict the risk beforehand, effectively control the risk in the process and effectively differentiate the risk afterwards through certain technical methods. If you want to make effective use of RF algorithm, you need to use orthogonal design method to make all parameters more reasonable. Compare the output data of RF algorithm with the real credit risk data, as shown in Figure 3.

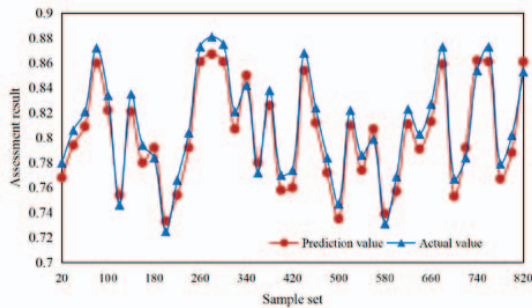


Figure 3. Machine learning results

It is not difficult to see that the learning result of RF algorithm is convergent, and it can approximate the original data well, so it has the basis of forecasting credit risk data. The indicators in the assessment model must be easily available. Generally speaking, the quantitative indicators should be obtained according to the balance sheet, profit and loss statement, cash flow statement and other statements

submitted by the loan application enterprise, while the qualitative indicators should be obtained according to the on-the-spot investigation of bank staff or the credit files established by the bank, otherwise the model will lose its practicability. The results of Internet credit risk assessment and expert assessment based on RF algorithm are shown in Table 1. It can be seen that the assessment results of training samples are in good agreement with those of experts.

TABLE 1. ASSESSMENT RESULTS OF EXPERTS AND RF ALGORITHM

Sample number	Expert appraisal	RF assessment
1	0.95	0.948
2	0.78	0.775
3	0.85	0.854
4	0.76	0.763
5	0.77	0.777
6	0.65	0.652
7	0.85	0.851
8	0.69	0.688
9	0.66	0.657
10	0.78	0.784

The purpose of bank loan risk scoring is to assess the credit status of the enterprise and judge the possibility of the bank's recovery. Therefore, the scoring result must be a comprehensive reflection of the enterprise's credit, that is, the index system should fully reflect the loan applicant's performance ability in social and economic activities, otherwise the result may be distorted and the actual use value will be small. The quality of business results is the foundation of an enterprise's existence and development. An enterprise with poor business performance is not only difficult to talk about development, but will be eliminated by the market sooner or later. Therefore, while analyzing the solvency of an enterprise, we should also observe its business situation and development prospects, which mainly include profitability and asset management capabilities. Comparing the prediction accuracy and average absolute error between the credit risk identification model in this paper and the traditional SVM, the results are shown in Figure 4 and Figure 5.

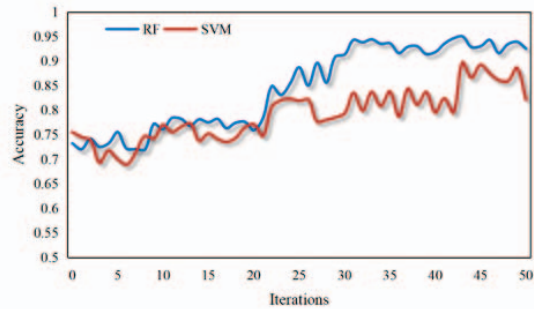


Figure 4. Accuracy comparison

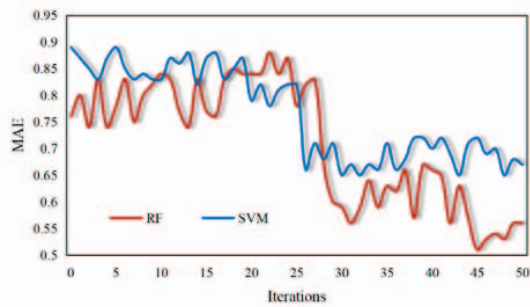


Figure 5. Comparison of average absolute error

As can be seen from Figure 4 and Figure 5, after many iterations, the accuracy of this method is obviously better than that of the traditional SVM algorithm, with the accuracy as high as 95.73%, 19.24% higher than that of SVM algorithm, and the error reduced by 30.26%. Therefore, the bank credit high-risk customer identification algorithm based on RF algorithm is a reasonable and feasible assessment model, which is of great significance to the construction of commercial bank market environment. Before the occurrence of credit risk, relevant personnel can carefully analyze and identify various risk factors in the process of credit management of commercial banks according to the nature and characteristics of credit risk, so as to provide the basis and conditions for credit risk control.

IV. CONCLUSIONS

Credit risk is influenced by national macro-economic policies, industry or regional economic development, micro-factors such as changes in customers' operating conditions, and meso-market factors. There are certain differences in the influence surface, degree and essence of various credit risk indicators on risk. To establish a risk early warning model, we can first study the relationship between several different types of variables and credit risks from several independent angles, and construct a relatively independent risk early warning subsystem. Then, considering the correlation among subsystems, the system is placed on the common platform of information collection, processing and application, and becomes a complete credit risk identification system. In this paper, a credit risk assessment model based on RF algorithm is proposed, and the initial RF is optimized by parameter optimization, which further improves the recognition accuracy of the model. After many iterations, the accuracy of this method is

obviously better than that of the traditional SVM algorithm, with the accuracy as high as 95.73%, 19.24% higher than that of SVM algorithm, and the error reduced by 30.26%.

Constructing the credit risk identification system of commercial banks with the help of modern intelligent algorithms can change the situation of superficial credit risk judgment and lagging risk response under the traditional management mode, and improve the technical content and accuracy of risk analysis. As for the classification of loan customers, we can make a detailed analysis of these customers by using the internal rating method for reference, and treat them differently in terms of the substantive sources, channels and differences of credit risks of each type of customers, so as to provide a more reliable basis for risk classification.

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